

Project specifications :

Supervised Trajectory analysis for a test of Optimal Foraging Theory

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Place : (Laboratory, Institute, Research Organism etc): Centre de Recherches sur la Cognition Animale,
Centre de Biologie Intégrative, CNRS

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behavioral ecology



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I. Context and objectives :

1) Biological context:

The long-term aim of the team is to adapt and test Optimal Foraging Theory to simple non-visual organisms, which comprise most of the tree of life (from microbes to lower invertebrates), and are nevertheless understudied from the point of view of Optimal Foraging. To do so, the lab use the nematode *Caenorhabditis elegans* as a powerful model system (1), and they focus on one of the keystones of Optimal Foraging Theory, the Marginal Value Theorem (2), which they have recently adapted to simple organisms (3). They have collected a rich dataset of the exploratory behavior of *Caenorhabditis elegans* in environments with regularly spaced food patches. This dataset consists of approximately 1500, 8 hour trajectories, under varying patch numbers and food densities. This dataset is ideally suited to test the Marginal Value Theorem.

2) Bioinformatics aspects and objectives :

The project is centered on the preprocessing and analysis of a large trajectory dataset using Python. All manual annotations and corrections have already been performed, and the data are ready for automated processing.

The first objective is to ensure data reliability by detecting missing, empty, or inconsistent files and generating a clean, well-documented trajectory database. The second objective is to perform initial behavioral analyses that will allow testing predictions of the simplified Marginal Value Theorem.

Specifically, the project aims to:

- Create a clean and reliable dataset by identifying empty videos and missing data.
- Automatically annotate and label each video according to data quality and biological relevance.
- Quantify spatial depletion patterns across food patches.

3) The Team :

The team of our Internship is : Interindividual Variability and Emergent Plasticity (IVEP). Lead by Jeanson Raphael, a researcher.

Our supervisor is Alfonso Pérez Escudero accompanied by Roger Lloret Cabot.

The IVEP team's study is the determinants of phenotypic plasticity across different biological scales, from cells to individuals and entire animal societies. Phenotypic plasticity, the ability of a genotype to produce different phenotypes in response to environmental variation, is one of the key mechanisms through which organisms adapt to changing conditions. Using an integrated and interdisciplinary approach, the scientists

are exploring how biological entities, be they cells, individuals or animal societies, react to and integrate environmental stimuli to produce adapted behavioral responses.

II. Functionalities:

This table lists all the functionalities, files, folders, scripts that we will use to complete the project aim:

Inputs files:	A folder : Minipatches_light_20260116 containing experimental folders (that represents various videos) that contain many files : - <code>traj.csv</code> : contains the trajectory data of detected worms over time. - <code>traj.mat</code> : contains manually reviewed and corrected food patch annotations. - <code>foodpatches_reviewed.mat</code> : contains manually reviewed and corrected food patch annotations. - <code>fodpatches.mat</code> : contains automatically detected food patches without manual correction. - <code>background.tif</code> : image that represents the static background of the experiment without worms. - <code>composite_patches.tif</code> : image that represents a composite visualization of all detected food patches over the experiment. - <code>holes.mat</code> : contains the positions of holes or invalid regions in the experimental arena. - <code>info_condition.mat</code> : stores metadata about the experimental conditions. - <code>maskROI.mat</code> : contains the Region Of Interest (ROI). - <code>silhouettes.mat</code> : contains worm silhouettes extracted from the video frames. - <code>times.mat</code> : contains timing information linking frame numbers to real experimental time. - A tool to visualize videos - A python class to extract the geometry of food patches
Output files:	A CSV file that compares data was created (results.csv), it contains the following columns: - Folder pat - Presence/absence of required files - File emptiness status - Diagnostic label for the video

	Later in the project, we will: • Create a script that does the analysis of multiple worms and produces a report • Create a script that does the analysis of speeds and produces a report • Create a script that produces a report about the holes, and fixes the ones that can be fixed • Erroneous cases will be visually confirmed using a video player to manually inspect the corresponding videos, the video will be constructed using all the available <code>.mat</code> files, which are combined to visualize the experiment and manually inspect tracking errors.
About Python	• Use of : ◦ librarys : pandas, os, pathlib ◦ Github ◦ VSCode

III. Aim of the project:

1) Step 1: Creation of a Clean Dataset :

a. Empty Videos or Missing Data

A video is considered inconsistent if:

- `traj.csv` is missing but `traj.mat` is not empty.
- `foodpatches_reviewed.mat` is missing but `traj.mat` is not empty.

A Python script loops through all folders and:

- Checks file existence.
- Measures file size to detect empty files.
- Labels each video as:
 - **OK**: all required files exist and are consistent.
 - **NO WORM**: no trajectory data (empty or missing `traj.mat`).
 - **WEIRD**: trajectory data exist but required files are missing.

All results are exported to an Excel/CSV file (Deliverable 3).

2) Step 2: Analysis of Multiple Worms :

For each video:

- Compute the proportion of frames containing multiple worms.
- Manually inspect selected ambiguous cases.
- Assign one of the following labels:
 1. Zero worms
 2. One worm, clean
 3. One worm with tracking errors (frames with multiple detections)
 4. Two or more worms

The number and indices of problematic frames are reported.

3) Step 3: Speed Analysis :

- Compute instantaneous speed along trajectories.
- Detect suspicious jumps indicating tracking errors or data corruption.
- Flag abnormal segments for later exclusion or correction.

4) Tools Provided :

The following tools are provided and not developed during this project:

- Video viewer
- Food patch class Various folders :

5) Distribution of the workload :

We distributed the workload evenly between the three group members. So we each had a third of the work to do. We worked together, using regular meet-ups at the university campus to progress through the project, specially on the python script and the report.

IV. Deliverables:

1. Python scripts for data cleaning and analysis
2. A clean, annotated trajectory dataset
3. A CSV/Excel file labeling all videos (Deliverable 3)
4. Clear documentation explaining:
 - Filtering criteria
 - File structure
 - Output interpretation

5. Python scripts to compute spatial patch depletion
6. A report summarizing the results on spatial patch depletion

V. Constraints/ Risks:

1) Functional Constraints :

- **Scientific Reproducibility:**
No data may be permanently deleted. All filtering decisions must be logged and reversible.
- **Validation Workflow:**
New filtering criteria must be validated by the supervisor before application to the full dataset.
- **Data Integrity:**
The association between trajectories and experimental conditions (patch number, food density) must be preserved.

2) Organizational Constraints :

- Fixed academic deadline aligned with the semester schedule.
- Limited team size requiring balanced task distribution.
- Regular weekly meetings with the supervisor to validate progress of the project.
- Use of shared tools (GitHub, google drive and docs and colab) for coordination.

3) Technical Constraints :

- Large dataset size requires efficient memory usage.
- Code must be portable across operating systems (Windows, Linux, macOS).
- Clear coding conventions and documentation must be followed.

Previsional calendar:

- Monday 12 jan : first meeting with our supervisor
- Tuesday 20 jan : 2nd meeting
- Wednesday 28 jan : 3rd meeting
- Monday 9 : 4th meeting

We intend to keep our weekly meetings going. So far we have finished the parts 1 and 2 (a and b):

1. Create auxiliary tools
 - a. Video viewer (Deliverable 1)
 - b. Foodpatch class (Deliverable 2)

2. Create a clean dataset
 - a. Empty videos or missing data
 - i. Folders without traj.csv should have an empty traj.mat
 - ii. Folders without foodpatches_reviewed.mat should also have empty traj.mat
 - iii. Do a script that checks this, reports the number of videos with these issues, and labels them (Deliverable 3)
 - b. Analysis of multiple worms
 - i. Compute the proportion of frames in which there are multiple worms.
 - ii. Analyze manually some cases, and annotate every video. Possible cases:
 1. Zero worms
 2. One worm clean (no frames with multiple worms)
 3. One worm with errors
 4. Report number of frames with errors, and which ones they are.
 5. Two (or more) worms.
 - iii. Suggest how to fix the erroneous multiple worms. Is there an automatic procedure to detect the false worm? If there are not too many, we can simply annotate them by hand.
 - iv. Create a script that applies these fixes and produces a report (Deliverable 4)
 - c. By the 4th week of February : Analysis of speeds
 - i. Find any suspicious jumps, and look at them. Are they tracking errors? Suggest a method to fix them. Should we exclude any videos because of this?
 - ii. Create a script that applies these fixes and produces a report (Deliverable 5)
 - d. By the 2nd week of march : Analysis of holes
 - i. Create a classification of holes, with the following categories:
 1. Fixable hole: The worm disappears and reappears almost at the exact same location.
 2. Edge excursion: The worm goes outside the tracked region and comes back (also near the edge).
 - a. Non-problematic: Outside the food patches.
 - b. Problematic: Inside a food patch.
 - c. Semi-problematic: The excursion does not touch any food patches, but some food patches overlap with the non-tracked region.
 3. Edge final excursion: The worm goes outside the tracked region and never comes back. It can also be problematic or non-problematic.
 4. Tracking mistake: The worm disappears and reappears, and it's not near the edge. We must first make a list of all these mistakes, and see how prevalent they are. Then, we must look at some of them manually to get a sense of what they look like. Some of them will not be too problematic, for example if it looks like the worm was outside the food patches all the time. Others will be more problematic. We need to decide how to handle them.
 - ii. Create a script that produces a report about the holes, and fixes the ones that can be fixed (Deliverable 6)

By the 4th week of march: Analyze spatial depletion in food patches

Executable Team Signature: _____

Supervisor Signature: _____

Date: _____